

Visualizing the Immune Mechanisms: Interactive Storytelling for Pediatric Allergy Engagement

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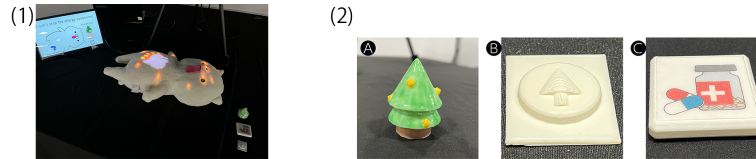


Figure 1: (1) Scene of Must cell character projected onto the soft toy's body. (2) Hospital Toy Devices. (A) simulating an allergen (B) simulating a tablet for SLIT (C) simulating a symptomatic antiallergic medications.

Abstract

This study presents an interactive media experience that uses medical play with a soft toy to explain allergic diseases and sublingual immunotherapy (SLIT) to children. By treating the soft toy, the experience supports children's understanding of disease mechanisms and treatment processes. A total of 34 children aged 3 to 11 participated. After the experience, participants completed a comprehension test and questionnaire. Results showed that the media effectively improved understanding of allergic diseases and SLIT. These findings suggest that the interactive experience can serve as an educational tool for pediatric patients undergoing long-term treatment.

CCS Concepts

• **Human-centered computing** → Mixed/augmented reality; Virtual reality; • **Applied computing** → Interactive learning environments.

Keywords

Children, Interactive art, Augmented reality, Assistive technology, Pediatric care, medical play, patient preparation, Alleviate anxiety, Hospitalized children's well-being, Child-centered design

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1 Introduction

Child Life Specialists (CLS) in pediatric care provide developmentally appropriate preparation and psychosocial support to help children cope with medical experiences [Boles et al., 2020; Romito et al., 2021]. Their work reflects the principles of Patient- and Family-Centered Care, which promotes shared decision-making through tailored information and emotional support [Committee on Hospital Care and Institute for Patient- and Family-Centered Care, 2012]. While hands-on methods such as medical play have proven effective, recent research has explored how digital and tangible media can further support children's comprehension. For example, Kyfonidis and Lennon developed an interactive tool that uses physical objects to educate children with type 1 diabetes [Kyfonidis and Lennon, 2019].

Inspired by the preparation strategies of CLS, this study develops an interactive educational artwork designed to help children's understanding of sublingual allergen-specific immunotherapy (SLIT), a long-term treatment for allergic diseases. Our artwork uses play-based interaction to explain the biological mechanisms of SLIT, fostering emotional engagement and a positive attitude toward treatment.

2 Implementation

SLIT is a form of allergen-specific immunotherapy involving the daily administration of dissolvable tablets under the tongue. Although clinically effective, SLIT adherence often declines over time [Okubo et al., 2023]. To address this, we designed an interactive experience based on consultations with pediatricians and nurses. Medical professionals emphasized that helping children grasp the purpose and flow of treatment is more valuable than detailed explanations to foster engagement with long-term therapy in a positive and emotionally supportive way. In response, we simplified the treatment process into three stages—much shorter than the actual

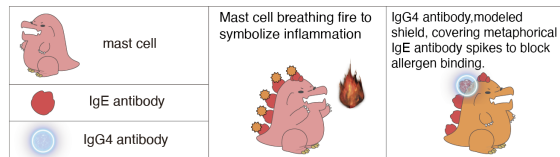


Figure 2: Design of mast cells, IgE, and IgG4

SLIT timeline—to support comprehension and emotional involvement.

The experience invites children to play the role of a doctor and administer treatment to a soft toy patient. By bringing specially designed toy instruments into contact with the toy’s mouth area, they trigger animated projections that visualize internal changes, progressing through the three treatment stages (Figure 1-(1)).

These instruments, referred to as Hospital Toy Devices, were 3D-printed and embedded with Near Field Communication (NFC) tags (Figure 1-(2)). An NFC reader in the toy detects the device and activates the corresponding animation projected onto the toy’s surface. A synchronized monitor provides visual prompts and explanations, supporting narrative immersion and understanding.

The story and characters were refined through feedback from children in multiple user testing sessions. To help them understand the biological processes behind allergic reactions, we focused on the immune mechanisms of “sensitization” and “provocation,” translating them into a narrative with characters modeled on mast cells, IgE antibodies, and IgG4 antibodies. For instance, mast cells were depicted as monster-like figures that breathe fire to symbolize inflammation (Figure 2). This storytelling metaphor helped children grasp causal relationships between immune responses and symptoms.

3 Consideration

We evaluated the artwork through an objective test and a subjective questionnaire, obtaining valid responses from 32 and 34 children aged 3–11. The test included multiple-choice questions with illustrations and text, designed to assess understanding of the disease, treatment content, and therapeutic effects. The questionnaire used a 5-point Likert scale to measure subjective impressions. As shown in Table 1, the accuracy rates were high across all test items, suggesting a solid understanding of the targeted medical concepts. Similarly, the questionnaire results (Table 2) show consistently high average scores, indicating that the experience was effective in promoting understanding of allergic diseases and SLIT, as well as in fostering a positive and engaging perception of treatment.

Healthcare professionals highlighted potential implementation challenges but also emphasized the experience’s value in helping children understand the purpose of treatment and in supporting their active participation—particularly in transitional care settings where greater independence is encouraged.

4 Conclusion

This study demonstrated that interactive media using medical play effectively enhanced children’s understanding of allergic diseases and SLIT, while fostering a more positive attitude toward treatment. These findings highlight the potential of interactive tools in pediatric education. Such approaches may complement traditional

Table 1: Results of the comprehension test

Question	Correct Answer Rate	SD
Understanding the mechanism of allergic reactions	94.1%	0.24
Understanding how to perform SLIT	76.5%	0.43
Understanding the effect of SLIT on mast cells	100%	0
Understanding the overall effect of SLIT upon treatment completion	97.1%	0.17

Table 2: Results of the post-experience questionnaire

Question	Mean Value	SD
Clarity of explanation regarding allergic disease	4.44	0.62
Clarity of explanation regarding the effects of SLIT	4.44	0.84
Positive attitude toward SLIT after experiencing the work	4.59	0.80
Engagement throughout the entire experience without losing interest	4.91	0.30
Enjoyment and engagement with the experience as a playful learning activity	4.94	0.25

explanations and apply to other conditions. While this study did not consider participants’ medical history or treatment adherence, future research should address these factors and examine long-term effects through follow-up studies.

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